

CIS 2103

Assurance, Security, and Privacy



كليات التقنية العليا
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Project Report

Big Data and Financial Privacy

Instructor	Dr. Nedaa Al Barghuthi
Student name	ID
Maryam Ali	H00535836
Sara Hussain	H00535574
Fatima Ali	H00535657
Mariam Yasir	H00534951
CRN	18083

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ABSTRACT

Big data analytics revolutionize the financial industry in terms of immense contributions to insight and service improvement. On the other hand, this technological development brings huge risks related to data privacy. Whereas financial institutions increasingly collect and analyze a large volume of personal and financial information, so does the concern over data breach, misuse, and unauthorized access. Deng et al. (2020)

Thus, public perception of privacy concerning data in the financial industry is felt through a survey by 106 individuals. The findings of the study present that big data is used, correspondingly the more privacy concerns are increasing. Much apprehension was expressed with regard to data breaches and access to data by third parties, misuse of personal data. In this respect, to win the confidence of customers, it is necessary to introduce transparency in the communication, as well as making the services secure. (Oyewole et al. 2024)

In this regard, every financial organization will have to adopt multiple ways of data privacy, including powerful encryption techniques for sensitive information, periodic security audits to search for and fix vulnerabilities, and transparent data practices that foster confidence in the customer. Most importantly, the need to sensitize employees on regulations and best practices pertaining to data privacy in order to avoid internal breaches.

It will aid the industry stakeholders in making pre-emptive efforts at protection from sensitive information risks and protect consumer confidence in a sea of constant change when it comes to data privacy in the financial sector. The study wraps up with suggestions for further research and innovation in overcoming the challenges presented by big data so that long-term security of financial data can be achieved.

INTRODUCTION

Financial institutions process a lot of sensitive personal and financial information. Because of this, they are one of the major sectors targeted in relation to cyberattacks. Regarding data privacy and information security, it is necessary for financial institutions to make the protection of customer data a priority to maintain the trust of customers. This includes informing all employees about the best security practices and data protection requirements through the implementation of tight security controls.

The purpose of this study is to analyze data privacy in relation to big data in the financial sector. Our research topic is "Data privacy in the era of big data in the financial sector," and we will examine the potential, difficulties, and best practices for protecting sensitive financial data.

We are hypothesizing that the big data analytics usage in the financial industry is increasingly developing great risks to data privacy that becomes a great concern for the client's personal data, especially in the financial sector; it is very important to apply privacy regulations and strong security measures to raise the security levels and to listen to the customer's feedback.

BACKGROUND

Recent discussions have pointed to increasing challenges and opportunities created by the intersection of big data and financial services. Research by Aldoush and Ferdous (2023) underlines the building of trust between financial institutions and their customers through ethical data practices. This is consistent with Chen et al.'s (2020) research, who purported measures in the field of security and following of regulations on protection of data. Meanwhile, Li et al. (2020) and Deng et al. (2020) examine the big data analytic potential within the context of the financial sector, considering the privacy risks inherently attached to such data. These combine to make a complex landscape in which innovation in financial technology has to balance the protection of customer privacy.

The article "The Financial Impact of Data Privacy Breaches" points out that even though the influence of cybersecurity breaches on the performance of an entity is investigated, the influences of data privacy breaches are not so well examined. Thus, Esty et al. (2020) conducted research on how such breaches affect firm value. Using a sample of 131 data privacy breaches in the U.S. from 2012 to 2014, this study examines the connections among firm attributes, attack information, and investor reactions. From these results, it follows that data privacy breaches negatively affect firm value; yet, for larger firms, the impact is less harmful compared with smaller firms. (Esty et al. (2020))

Deng et al. (2020) say that big data analytics is increasingly applied within the financial sector to come up with value-added insights and improve services. The trend, growing at an increasing rate, is underpinning serious data privacy concerns. The authors indicate that the breach of information and misuse of data might shake users trust in financial stability. They therefore call for improved security measures and strict protection of customer data. Deng et al. (2020) further emphasize finding an appropriate balance in the exploitation of big data for financial development while ensuring security in sensitive customer information.

Data privacy regulations are evolving rapidly to keep pace with the growth of digital financial services, aiming to protect customer data in today's data-driven economy. With more people using digital tools like online payments, peer-to-peer lending, and wealth management platforms, there's a growing need for tougher regulations to ensure consumer safety and privacy. The EU's GDPR has set a widely accepted standard for data privacy, pushing many countries to align their own regulations with its strong protections. While this global trend helps build customer trust and makes companies more competitive, it can be challenging for financial

technology businesses that must navigate different laws in each country they operate in. (Oyewole et al. (2024))

As digital financial services continue to evolve with new innovations, data privacy regulations are also adapting to find a balance—one that protects people's personal information while supporting tech-driven progress in this rapidly growing sector. (Oyewole et al. (2024))

Despite all that said above, threshold cryptography ensures that your data is protected and would remain confidential, unavailable to those who are unauthorized. Threshold cryptography is a cryptographic scheme where a cryptographic key is divided into several shares and distributed among different parties. In order to decrypt it, you have to bring these parts together and perform different types of operations to get the key. This, in turn, makes things more secure because no one person or organization has complete access to credential information. Each of the custodians holds only a share of the key, not the whole key. The immediate advantages of threshold cryptography are enhanced security and strength, along with shared responsibility. Not even in cases of compromise would the key be rebuildable without the minimum number of shares, thus reducing insider threats and unauthorized access. (Krishnasamy, 2024)

METHODOLOGY

Our research focuses on data privacy in the era of big data and data privacy within the financial sector. We asked people a couple questions about financial privacy and how aware they are of it. We asked people to participate voluntarily, reminding them that their information would remain confidential, alongside their identity to remain anonymous.

To collect data, we conducted a survey (Big Data and Financial Privacy Survey, 2024) targeting approximately 100 participants, designed to gauge their perspectives and understanding of data privacy issues. The survey comprised 12 questions: 4 yes-or-no questions, 3 agree/disagree statements, 3 scale-based questions ranging from "never" to "always," and 2 open-ended questions to gather deeper insights into participants' thoughts and attitudes, providing valuable qualitative feedback.

The population for this survey included friends, family, and colleagues, resulting in a sample size of 106 responses, with a response rate of 53%. We employed a random convenience sampling method, selecting participants based on ease of access and availability. Although this method limits the background information of our respondents, it allowed us to gather an understanding of public perceptions on data privacy in the financial sector. The combination of both quantitative and qualitative responses enabled us to analyze trends and personal insights, enriching our findings.

Parameter	Description
Population of the study	Family, friends, and colleagues
Sample size	106 responses
Sampling method	Random convenience sampling
Response rate	$106/200 \times 100 = 53\%$

FINDINGS

Our survey gathered 106 responses through a set of 12 thoughtfully designed questions aimed at understanding perspectives on data privacy in the financial sector. The questionnaire incorporated a variety of formats, including yes-or-no options, agree-to-disagree statements, scaled responses ranging from "never" to "always," and open-ended short-answer questions to capture different insights. (Big Data and Financial Privacy Excel, 2024)

We began by collecting demographic data on gender and age. 75% of the responders were between the ages of 18 and 25, and 81% were female. Around 20% were above 25, with only a small portion of 5% under 18. This demographic data helped us understand the primary makeup of our sample. (Big Data and Financial Privacy Excel, 2024)

The significance of data privacy was covered in the first set of questions. Almost all participants who were asked if they thought data privacy was important in the banking sector said "yes," with only two out of 106 responding negatively. However, 58% of people who replied to our survey said they were uncomfortable about sharing personal information with financial institutions and 42% said they were somewhat comfortable doing so. By asking respondents if they were aware of how these institutions handled their data, we were able to further investigate awareness. There was a nearly equal distribution of replies, with 52 respondents expressing some knowledge and 54 indicating little awareness (see Figure 1). When asked if they had previously chosen not to share more information with financial institutions, 57% of respondents claimed they had, whereas 43% said they had not. (Big Data and Financial Privacy Excel, 2024)



Figure 1

We investigated respondents' opinions of financial institutions' attempts to protect data in the agree-to-disagree section. 44 respondents disagreed with the statement that financial institutions are doing enough to protect client data, implying that the procedures in place are not enough. Just 24 respondents agreed that institutions are sufficiently securing the data of their consumers, while 38 respondents had no view. (Big Data and Financial Privacy Excel, 2024)

Next, we examined views on whether the use of big data improves customer service but compromises privacy. 40 respondents agreed that big data enhances service at the cost of privacy, while 34 disagreed and 32 remained neutral. (Big Data and Financial Privacy Excel, 2024)

Finally, we asked about concerns regarding potential misuse of data by financial institutions. The majority, 45 respondents, expressed concern, agreeing that misuse is a possibility. 35 disagreed with this sentiment, and 26 had no opinion. These responses indicate a prevalent unease about privacy practices in the financial sector, with significant skepticism regarding data protection measures and the ethical use of personal data by financial institutions. (See Figure 2 below). (Big Data and Financial Privacy Excel, 2024)

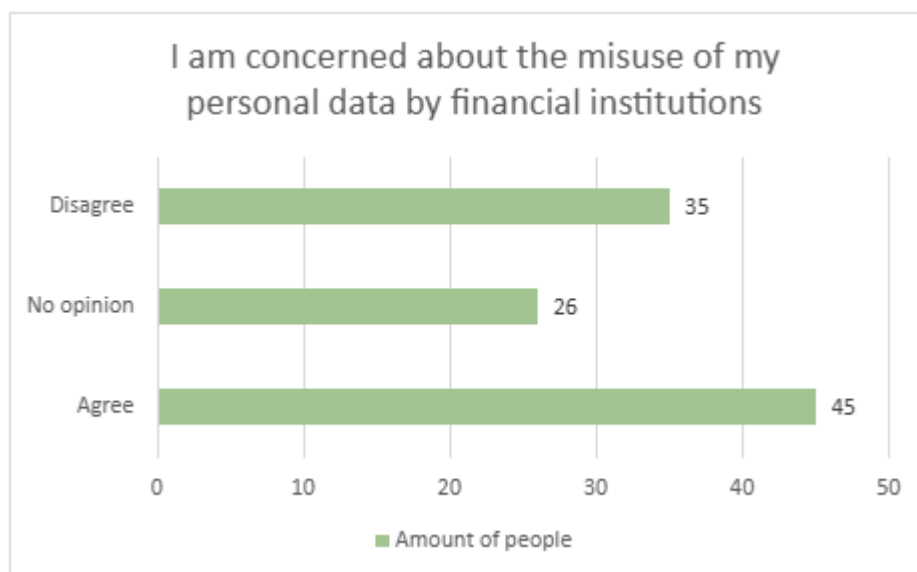


Figure 2

In the scale-based questions (from "never" to "always"), we first asked participants how often they review the privacy policies of financial institutions before sharing their data. 15.1%

responded "never," 27.4% chose "rarely," and 35.8% selected "sometimes" being the most common response. Meanwhile, 11.3% said "often," and 10.4% responded "always."

We then asked how frequently participants worry about their personal financial data being compromised (see Figure 3 below). The responses were 1.9% "never," 13.2% "rarely," with 40.6% being the largest group and choosing "sometimes." Additionally, 24.5% selected "often," and 19.8% indicated "always." (Big Data and Financial Privacy Excel, 2024)

Finally, we asked how frequently individuals use two-factor authentication or change their passwords to protect their private financial information. Among those surveyed, 3.8% selected "never," 17.9% chose "rarely," 41.5%, the largest group opted for "sometimes," 27.1% responded with "often," and 15.1% indicated "always." These results highlight that while some participants actively take steps to secure their data, a substantial portion engages in such protective measures infrequently or sporadically. (Big Data and Financial Privacy Excel, 2024)

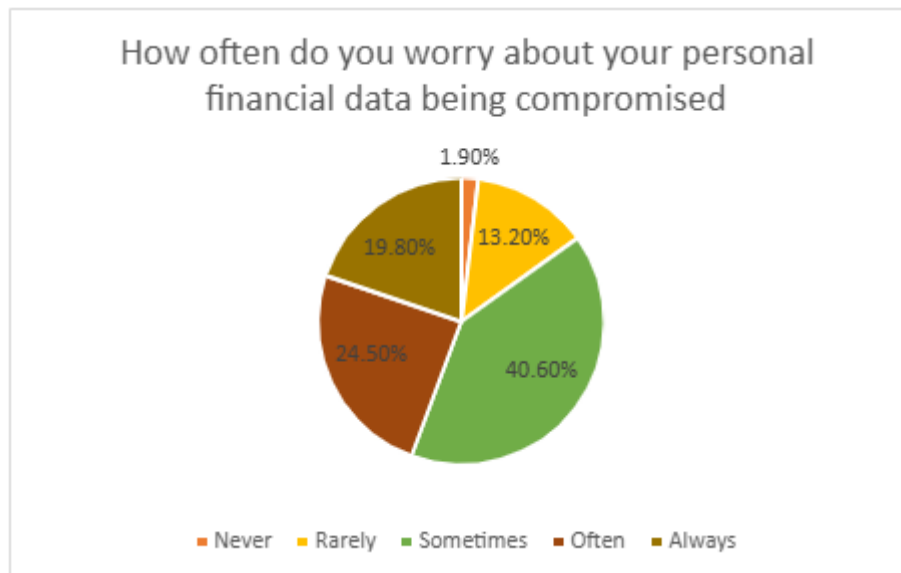


Figure 3

We included open-ended questions at the end of the poll to get further opinions and suggestions from respondents. Data breaches were cited as the top concern when questioned about the financial industry's main data privacy problems. Many respondents were concerned that their private information would be compromised, which might have serious repercussions like identity theft, financial loss, and illegal access. Furthermore, a sizable portion raised issues with third-party access, particularly the dangers of financial institutions exchanging data with other parties. Mostly if third-party partners do not adhere to the same privacy requirements as the banking institution, there was concern that data would be sold or shared without consent.

Respondents also noted potential human errors, system flaws, insider threats, and insufficient data anonymization as additional risk factors, emphasizing that even small lapses could result in substantial privacy issues. (Big Data and Financial Privacy Excel, 2024)

In response to the question, “What should financial institutions do to improve data privacy in the era of big data?” Several key ideas were brought to our attention. Many participants advocated for the implementation of advanced encryption and recommended regular updates and improvements to security infrastructure. Another prominent theme was the need for greater transparency and communication between financial institutions and customers. Respondents emphasized that institutions should clearly explain how data is collected, used, stored, and shared. (Big Data and Financial Privacy Excel, 2024)

Moreover, participants highlighted the importance of employee training on data privacy practices to prevent internal data misuse. Some also suggested the use of artificial intelligence and machine learning to detect and prevent potential data breaches proactively. Overall, these insights indicate that customers seek both robust technical safeguards and transparent, customer-centred policies to feel secure and confident in sharing their data with financial institutions. (Big Data and Financial Privacy Excel, 2024)

DISCUSSION

The results of our survey (Big Data and Financial Privacy Survey, 2024) strongly back up our hypothesis, offering evidence that validates our assumptions and the key points of our research. The research aligns with the CIA (Confidentiality, Integrity, Availability) triad, demonstrating the critical aspects of data privacy in the financial sector. These findings further complement the insights from the literature review, showcasing a consistent relationship between big data analytics and rising privacy concerns.

Our results align with the research of Chen et al. (2020), Aldoush and Ferdous (2023), which talks about the importance of following legal regulations and implementing secure data handling practices; customer trust is of most importance and maintaining it should be their top priority. These studies reinforce the idea that prioritizing transparency and compliance is key to fostering confidence in financial institutions. For instance, 58% of respondents in our survey expressed discomfort with sharing personal information with financial institutions, highlighting a significant lack of trust in how data is managed. (Big Data and Financial Privacy Excel, 2024) The same discomfort is shown by Deng et al. (2020) and Li et al. (2020), who described how privacy risks in big data analytics can undermine user trust in the financial sector.

Our respondents' main concern revolved around confidentiality, particularly regarding data breaches, unauthorized access, and illegal sharing of personal information. This aligns with findings done by Esty et al. (2020), who highlighted the damaging effects of data breaches on company value, especially for smaller enterprises. Our survey participants mentioned fears of identity theft, financial loss, and third-party misuse of personal data, continuously reminding us of how breaches impact not only customer trust but also business sustainability.

Issues of integrity were evident, as 54% of our responses said they were unaware of how financial institutions handle their data. This lack of transparency raises concerns about the potential risk of data misuse or alteration without clients' knowledge. Participants kept bringing up the need for financial institutions to adopt transparent and honest communication regarding data handling practices to gain trust and confidence among customers.

Availability was brought up by respondents highlighting the importance of regular system updates, advanced encryption, and great security protocols to ensure data remains secure and accessible even during cyberattacks or system failures. The demand for such measures demonstrates a clear expectation for financial institutions to prioritize strong technical safeguards to protect sensitive information.

The survey's feedback also reflects a demand for legal structure, such as GDPR, to reduce privacy risks in digital finance ecosystems. Participants suggested more transparent data usage policies and frequent security updates. Oyewole et al. (2024), talks about the importance of protecting consumer data during an era of quick digital transformation.

One drawback of our research was our demographic of the sample, which consisted mostly of younger people between 18-25 and mostly females (81%). Younger populations may have different perspectives on data privacy compared to older demographics. Additionally, the survey questions focused primarily on perceptions of privacy but lacked deeper exploration into technical or industry-specific knowledge. The absence of targeted respondent groups, such as finance or IT professionals, further limits the scope of findings.

To address these limitations, future research could broaden the demographic scope by including a more diverse age range and professional backgrounds in finance and technology. Incorporating case studies on actual financial data breaches could provide practical insights into the effectiveness of existing privacy measures. Additionally, exploring privacy-preserving technologies like threshold cryptography (Krishnasamy, 2024) could provide new perspectives on securing the CIA triad and addressing customer concerns effectively.

CONCLUSION

Our research gives additional evidence that, while big data analytics have value for the financial industry, they also come with many significant privacy risks as confirmation of our hypothesis and findings. With issues revolving around data breach, third-party sharing, this puts a critical policy need for effective data protection, practice of transparency, and strict security.

This will hopefully form a core approach by which financial institutions can effectively minimize privacy risk. This addresses important building blocks in keeping the trust of the customers, sticking to changing regulations, and ensuring that sensitive financial information is not harmed in this completely digital world.

All of them use big data to enhance services, so financial institutions have to balance innovation and customer privacy. Whereas big data provides improved services to the appreciative people, a lot of concern has been raised on the use and protection of personal information. Many customers are afraid of how their data is treated and whether financial institutions take enough security measures. This implies the need for much stronger encryption, clear communication of data practices, and strong protection of privacy to guarantee security and confidence among customers.

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